Title: - Predicting the Habitability of Exoplanets Using Machine Learning

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🌌 ExoHabit – Exoplanet Habitability Predictor

🚀 1. Project Overview

ExoHabit is an **AI-powered system** designed to predict whether an **exoplanet can support life**.

It uses a **Machine Learning model (XGBoost)** to analyze planetary and stellar data and provides:

- ✅ Habitability Prediction (Habitable / Not Habitable)
- 📊 Confidence Score
- 🏆 Habitability Ranking among planets

🧠 2. Core Concept

The ExoHabit system is based on the idea that a planet's ability to support life depends on a combination of several important **astrophysical factors**. Instead of relying on a single parameter, the system analyzes multiple conditions together to make an accurate prediction.

- 🌍 Planet size and mass
- 🌡️ Temperature conditions
- ☀️ Distance from the star
- ⭐ Star characteristics

🌍 Planet Size and Mass

The size and mass of a planet play a crucial role in determining habitability.

- If a planet is **too small**, it cannot hold an atmosphere, which is essential for life.

- If it is **too large**, it may become a gas giant with no solid surface.
  👉 Earth-like planets with moderate size and mass are considered most suitable for life.

🌡️ Temperature Conditions

Temperature determines whether liquid water can exist on the planet.

- Extremely **high temperatures** can evaporate water.
- Extremely **low temperatures** can freeze it.
  👉 A balanced temperature range is required for sustaining life.

☀️ Distance from the Star

The distance between the planet and its star (called the **habitable zone**) is very important.

- Too close → Planet becomes too hot
- Too far → Planet becomes too cold
  👉 The correct distance allows the presence of liquid water.

⭐ Star Characteristics

The type and nature of the host star also affect habitability.

- Stable stars provide consistent energy
- Very hot or unstable stars may harm life
  👉 K-type stars (orange dwarfs) are often considered good candidates for supporting habitable planets.

These inputs are processed using a trained ML model to generate predictions.

⚙️ 3. How the System Works (Step-by-Step)

🧠 Step 1: Input Data

User provides planetary and stellar parameters:

🌌 Planetary Parameters

- Planet Radius → **2.759 Earth radii**
- Planet Mass → **6.63 Earth masses**
- Orbital Period → **9.15 days**
- Semi-Major Axis → **0.0814 AU**
- Equilibrium Temperature → **828 K**
- Density → **1.73 g/cm³**

☀️ Stellar Parameters

- Stellar Temperature → **5320 K**
- Luminosity → **0.519 (Sun = 1)**
- Metallicity → **-0.02**
- Star Type → **K-type (Orange dwarf)**

⚙️ Step 2: Machine Learning Processing

- Input data is passed to the **XGBoost model**
- Model compares it with trained dataset
- Identifies patterns of habitable planets

📊 Step 3: Output Result

System generates:

- ✅ **Prediction:** Potentially Habitable
- 📈 **Habitability Score:** 100%
- 🎯 **Confidence:** 100%
- 🔢 **Raw Score:** 0.999990

👉 **Meaning:** Very high probability of supporting life

🔷 4. What is Habitability Score?

- It is a value between **0 and 1**
- Closer to **1** → **More habitable**

Example:

- **0.999990 ≈ Highly habitable**

🧮 5. Habitability Score Explanation

The score is calculated based on:

- 🌍 Distance from star (Habitable Zone)
- 🌡️ Temperature suitability (not too hot / cold)
- 🪨 Planet composition (rocky preferred)
- ⭐ Star stability

👉 Score Range

- **0 – 40% → Not Habitable**
- **40 – 70% → Possibly Habitable**
- **70 – 100% → Highly Habitable**

🏆 6. Habitability Rankings

The system ranks exoplanets based on predicted scores.

🔝 Top Results

- Top 10 planets have ~**99.999% scores**

📌 Key Observations

- Radius → **2–3 Earth radii**
- Temperature → **400K – 800K**
- Stellar Temperature → **4200 – 5400 K**

👉 **Insight:**
Planets with **balanced temperature, moderate size, and stable stars** rank highest.

🔄 7. System Workflow

1. User enters planetary & stellar parameters
2. Data preprocessing & normalization
3. Features passed into trained **XGBoost model**
4. Model predicts:
    - Class (Habitable / Not Habitable)
    - Probability Score
5. Results displayed with ranking

🔷 8. Technologies Used

- 💻 Machine Learning: XGBoost
- 🧠 AI Model Training
- 🌐 Frontend: Web Interface
- 📊 Data: Exoplanet dataset (NASA-like data)

🔷 9. Advantages of ExoHabit

✔ Fast prediction
✔ Accurate ML-based results
✔ User-friendly interface
✔ Helps in space research

🔷 10. Limitations

❌ Depends on dataset quality
❌ Cannot guarantee real life presence
❌ Assumes Earth-like life conditions

🔷 11. Future Improvements

- Add **real-time NASA data API**
- Improve model accuracy with deep learning
- Include atmosphere & water detection
- 3D visualization of planets

🔷 12. Conclusion

ExoHabit is an innovative AI system that:

- Predicts habitable planets
- Uses machine learning for space science
- Helps humans explore possibilities of life beyond Earth 🌍

Link of website: - https://inspiring-haupia-ea27aa.netlify.app/